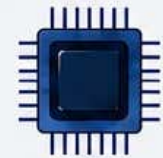
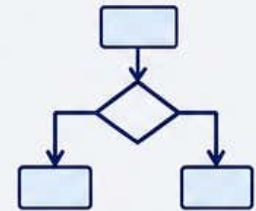




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# Syllabus



## Major Course (Elective-2): Machine Learning

| Level | Semester | Course Code | Course Name             | Credits | Teaching Hours | Exam Duration | Max Marks |
|-------|----------|-------------|-------------------------|---------|----------------|---------------|-----------|
| 5.5   | VI       | 112226      | <b>Machine Learning</b> | 2       | 30             | 2 Hrs         | 30        |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                          |                                    |                                                      |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------|------------------------------------------------------|
| <b>Course Objectives:</b> | <ol style="list-style-type: none"><li>1. To introduce students to the basic concepts, types, applications, and workflow of Machine Learning.</li><li>2. To develop an understanding of data preprocessing techniques required for preparing datasets for Machine Learning models.</li><li>3. To enable students to understand and apply supervised learning techniques for regression and classification problems.</li><li>4. To introduce students to unsupervised learning techniques, dimensionality reduction, and basic handling of image data.</li><li>5. To provide basic knowledge of neural networks and model evaluation techniques used to measure the performance of Machine Learning models.</li></ol> |                          |                                    |                                                      |
| <b>Course Outcomes:</b>   | <ol style="list-style-type: none"><li>1. Students will be able to explain the basic concepts of Machine Learning and apply preprocessing techniques to prepare data for model training.</li><li>2. Students will be able to apply regression and classification algorithms such as Linear Regression, Logistic Regression, KNN, Decision Tree, and SVM.</li><li>3. Students will be able to analyze clustering methods, dimensionality reduction using PCA, and understand the basics of image data processing.</li><li>4. Students will be able to describe basic neural network concepts and evaluate Machine Learning models using standard performance measures.</li></ol>                                      |                          |                                    |                                                      |
| <b>Unit System</b>        | <b>Contents</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Workload Allotted</b> | <b>Weightage of Marks Allotted</b> | <b>Incorporation of Pedagogies</b>                   |
| <b>Unit I</b>             | <b>Introduction &amp; Data Preprocessing</b> Definition of Machine Learning. Traditional programming vs. ML. Application of Machine Learning, Types of ML: Supervised, Unsupervised, and Reinforcement Learning. Machine Learning pipeline: data collection, preprocessing, training, testing, evaluation and prediction. <b>Data Preprocessing:</b> Handling missing values, categorical data encoding, feature scaling, and train-test splitting. <b>Preprocessing of Image Data:</b> image resizing, grayscale conversion, converting images or video frames into arrays, pixel normalization.                                                                                                                   | 8 Hrs                    | 8 Marks                            | Use any pedagogical technique suitable for the topic |
| <b>Unit II</b>            | <b>Supervised Learning Regression:</b> Simple and Multiple Linear Regression, Cost function, basics of Gradient Descent.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 7 Hrs                    | 7 Marks                            |                                                      |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |         |  |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------|--|
|                    | <b>Classification:</b> Logistic Regression, K-Nearest Neighbours (KNN), and Decision Trees. Naive Bayes Classifier, Support Vector Machines (SVM)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |         |  |
| <b>Unit III</b>    | <b>Unsupervised Learning:</b> Introduction, clustering, K-Means clustering algorithm, choosing number of clusters using Elbow Method.<br><b>Dimensionality Reduction:</b> Need, introduction to Principal Component Analysis(PCA).<br><b>Image Feature Extraction:</b> Basics of image data, converting images into arrays, resizing images, extracting basic features from image data for classification models.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 8 Hrs | 8 Marks |  |
| <b>Unit IV</b>     | <b>Neural Networks &amp; Model Evaluation</b><br><b>Neural Networks:</b> Biological vs. Artificial Neurons, the Perceptron model, basic activation functions (Sigmoid, ReLU).<br><b>Evaluation:</b> Confusion Matrix, Accuracy, Precision, F1-Score, Recall. Understanding Underfitting, Overfitting, and using Cross-validation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 7 Hrs | 7 Marks |  |
| <b>References:</b> | <b>Course Material/Learning Resources</b><br><br><b>Text books:</b> <ol style="list-style-type: none"> <li>1. Tom M. Mitchell, Machine Learning, McGraw-Hill.</li> <li>2. Ethem Alpaydin, Introduction to Machine Learning, MIT Press.</li> <li>3. Muller, Andreas C. and Guido, Sarah. Introduction to Machine Learning with Python: A Guide for Data Scientists. O'Reilly Media</li> <li>4. Theobald, Oliver. Machine Learning for Absolute Beginners: A Plain English Introduction. 3rd Edition, Scatterplot Press, 2021.</li> </ol> <b>Reference Books:</b> <ol style="list-style-type: none"> <li>1. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer.</li> <li>2. Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, An Introduction to Statistical Learning, Springer. (<a href="#">Springer</a>)</li> <li>3. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning, Springer.</li> <li>4. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow, O'Reilly Media.</li> <li>5. Sebastian Raschka and Vahid Mirjalili, Python Machine Learning, Packt Publishing.</li> <li>6. Kevin P. Murphy, Machine Learning: A Probabilistic Perspective, MIT Press.</li> </ol> <b>Weblink to Equivalent MOOC on SWAYAM if relevant:</b> <a href="https://onlinecourses.swayam2.ac.in/e-learning/preview/imb26_mg46">https://onlinecourses.swayam2.ac.in/e-learning/preview/imb26_mg46</a><br><a href="https://onlinecourses.swayam2.ac.in/e-learning/preview/ini26_cs01">https://onlinecourses.swayam2.ac.in/e-learning/preview/ini26_cs01</a><br><a href="https://www.coursera.org/specializations/machine-learning-introduction?utm_source">https://www.coursera.org/specializations/machine-learning-introduction?utm_source</a><br><a href="https://nptel.ac.in/courses/106106139?utm_source">https://nptel.ac.in/courses/106106139?utm_source</a><br><a href="https://nptel.ac.in/courses/106105152?utm_source">nptel.ac.in/courses/106105152?utm_source</a><br><a href="https://developers.google.com/machine-learning/crash-course?utm_source">https://developers.google.com/machine-learning/crash-course?utm_source</a><br><a href="https://openlearninglibrary.mit.edu/courses/course-v1%3AMITx%2B6.036%2B1T2019/course/?utm_source">https://openlearninglibrary.mit.edu/courses/course-v1%3AMITx%2B6.036%2B1T2019/course/?utm_source</a> |       |         |  |

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|  | <p><b>Web link to Equivalent Virtual Lab if relevant:</b><br/> <a href="https://ml-iitr.vlabs.ac.in/">https://ml-iitr.vlabs.ac.in/</a><br/> <a href="https://ml-dei.vlabs.ac.in/">https://ml-dei.vlabs.ac.in/</a><br/> <a href="https://cse19-iiith.vlabs.ac.in/">https://cse19-iiith.vlabs.ac.in/</a></p> <p><b>Digital Resources</b></p> <ol style="list-style-type: none"> <li>1. <a href="https://www.geeksforgeeks.org/machine-learning/machine-learning/">https://www.geeksforgeeks.org/machine-learning/machine-learning/</a></li> <li>2. <a href="https://www.kaggle.com/code/abdallahwagih/machine-learning-notes">https://www.kaggle.com/code/abdallahwagih/machine-learning-notes</a></li> <li>3. <a href="https://learnwith.campusx.in/products/100-Days-of-Machine-Learning-Notes-663a0d6101204179049a591b">https://learnwith.campusx.in/products/100-Days-of-Machine-Learning-Notes-663a0d6101204179049a591b</a></li> <li>4. Scikit-learn User Guide. (Scikit-learn)</li> <li>5. Scikit-learn Supervised Learning Documentation. (Scikit-learn)</li> </ol> |
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### Assessment Rubric: Internal for Theory Course (Major Elective-2)

| SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI<br>THEORY EXAMINATION<br>B.Sc. III, SEMESTER – VI (NEP) |                                                                                                                                                                                                                                                                                     |                                       |                      |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------|
| <b>Theory</b>                                                                                         | <b>Course Code:</b> 112226                                                                                                                                                                                                                                                          | <b>Course Title:</b> Machine Learning | <b>Max Marks:</b> 20 |
| <b>Sr. No.</b>                                                                                        | <b>Assessment Criteria</b>                                                                                                                                                                                                                                                          |                                       | <b>Marks</b>         |
| <b>1</b>                                                                                              | Any Two of the Activity (weekly or monthly assignment/ Unit Test/Class Test) <b>(CAT-1)</b>                                                                                                                                                                                         |                                       | <b>10</b>            |
| <b>2</b>                                                                                              | Any One of the Activity includes Open Book Test, MCQ Tests, / Theory related Class activity/Task etc.) etc. <b>(CAT-2)</b>                                                                                                                                                          |                                       | <b>5</b>             |
| <b>3</b>                                                                                              | Participation in any one Activity includes Quiz, Seminar, Group Discussion, Industry visit, Case Study, Mini project, innovative work, etc. based on Theory Course. Involvement in subject related activities such as Conference, Workshop, Coding Competitions etc. <b>(CAT-3)</b> |                                       | <b>5</b>             |